Listing 13.3 Finding a device by ID

```
static Optional<Device> find(Database database, String
deviceId) {
    return database.findOptional(Device.class,    #A
            "SELECT device_id, manufacturer, model,
encrypted_psk " +    #B
                    "FROM devices WHERE device_id = ?",
deviceId);    #C
}
```

#A Use the findOptional method with your Device class to load devices
#B Select device attributes in the same order they appear in the constructor
#C Use a bind parameter to query for a device with the matching device_id

Now that you have some device details, you can use them to authenticate devices and perform access control based on those device identities, which you'll do in section 13.1.2 and 13.1.3.

13.1.2 Device certificates

An alternative to storing device details directly in a database is to instead provide each device with a certificate containing the same details, signed by a trusted certificate authority. Although traditionally certificates are used with public key cryptography, you can use the same techniques for constrained devices that must use symmetric cryptography instead. For example, the device can be